Year 10 Mathematics

Number 3

6.1 Laws of Indices

Laws of Indices

STARTER · 5 MIN

Date: _____ **Title:** Laws of Indices

LEARNING OBJECTIVES

- Multiply and divide powers of the same base
- Raise a power to a power
- Use the zero index and simplify expressions with indices

STARTER *Answer in your book. No calculators.*

1. Work out 2^5

2. Work out 3^4

3. Work out 10^3

4. Work out $5^2 \times 2^3$

Starter — Answers

STARTER

Tick or correct in green pen

1. $2 \times 2 \times 2 \times 2 \times 2 = \mathbf{32}$
2. $3 \times 3 \times 3 \times 3 = \mathbf{81}$
3. $10 \times 10 \times 10 = \mathbf{1000}$
4. $25 \times 8 = \mathbf{200}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can multiply and divide powers of the same base	☐
SILVER	I can raise a power to a power	☐
GOLD	I can use the zero index and simplify expressions with indices	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Laws of Indices

INTRODUCTION · 3 MIN

The index (power) tells you how many times to multiply the base by itself.

Multiply

Add the indices.

$a^m \times a^n = a^{m+n}$

Divide

Subtract the indices.

$a^m \div a^n = a^{m-n}$

Power of a power

Multiply the indices.

$(a^m)^n = a^{mn}$

Zero index

Anything to the power 0 is 1.

$a^0 = 1$

Examples 1 and 2 — Multiplying and Dividing

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint $a^m \times a^n = a^{m+n}$ $\quad a^m \div a^n = a^{m-n}$

Example 1

Simplify, leaving your answer as a power:

$5^4 \times 5^3$

Example 2

Simplify:

$x^9 \div x^4$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$5^4 \times 5^3 = 5^{4+3}$

$= 5^7$

Tip: same base: add the indices.

Example 2

$x^9 \div x^4 = x^{9-4}$

$= x^5$

Tip: same base: subtract the indices.

Task 1

PRACTICE · 9 MIN

BRONZE

Simplify, leaving each answer in index form

Hint $a^m \times a^n = a^{m+n}$ $\quad a^m \div a^n = a^{m-n}$

1 $3^5 \times 3^2$

2 $7^8 \div 7^3$

3 $a^4 \times a^6$

4 $y^{10} \div y^2$

5 $2^3 \times 2^4 \times 2$

★ $m^5 \times m^4 \div m^3$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 $3^{5+2} = \mathbf{3^{7}}$

2 $7^{8-3} = \mathbf{7^{5}}$

3 $a^{4+6} = \mathbf{a^{10}}$

4 $y^{10-2} = \mathbf{y^{8}}$

5 $2^{3+4+1} = \mathbf{2^{8}}$

★ $m^{5+4-3} = \mathbf{m^{6}}$

Examples 3 and 4 — Powers of Powers and Zero

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint $(a^m)^n = a^{mn}$ $a^0 = 1$

Example 3

Simplify:

$(x^3)^4$

Example 4

Simplify:

$(2a^3)^2$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$(x^3)^4 = x^{3\times 4}$

$= x^{12}$

Tip: power of a power: multiply.

Example 4

$(2a^3)^2 = 2^2 \times a^{3\times 2}$

$= 4a^6$

Tip: square the number as well.

Task 2

PRACTICE · 9 MIN

SILVER

Simplify each expression

Hint $(a^m)^n = a^{mn}$ $a^0 = 1$

1. $(5^2)^3$
2. $(y^4)^5$
3. 7^0
4. $(3x^2)^3$
5. $x^5 \times x^0$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $5^{2\times3} = \mathbf{5^{6}}$

2. $y^{4\times5} = \mathbf{y^{20}}$

3. $7^{0} = \mathbf{1}$

4. $3^{3} \times x^{2\times3} = \mathbf{27x^{6}}$

5. $x^{5+0} = \mathbf{x^{5}}$

Examples 5 and 6 — Simplifying Expressions

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint $a^m \times a^n = a^{m+n}$ $\quad a^m \div a^n = a^{m-n}$

Example 5

(2 marks)

Simplify fully:

$3a^4b^2 \times 5a^3b$

Example 6

(2 marks)

Simplify fully:

$$\frac{12x^7y^5}{4x^2y}$$

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$3 \times 5 \times a^{4+3} \times b^{2+1}$

$= 15a^{7}b^{3}$

Tip: numbers multiply, indices add.

Example 6

$12 \div 4 \times x^{7-2} \times y^{5-1}$

$= 3x^{5}y^{4}$

Tip: numbers divide, indices subtract.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint $a^m \times a^n = a^{m+n}$ $\quad a^m \div a^n = a^{m-n}$

1. Simplify $4x^3 \times 6x^5$ (2 marks)

2. Simplify $\dfrac{20a^8}{5a^3}$ (2 marks)

3. Simplify $(2y^4)^3$ (2 marks)

4. Simplify $\dfrac{6p^5q^3 \times 2pq^2}{4p^2q}$ (3 marks)

5. Find n: $3^n \times 3^4 = 3^{10}$ (2 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1 $24x^{3+5} = \mathbf{24x^{8}}$

2 $4a^{8-3} = \mathbf{4a^{5}}$

3 $2^{3}y^{12} = \mathbf{8y^{12}}$

4 $\dfrac{12p^{6}q^{5}}{4p^{2}q} = \mathbf{3p^{4}q^{4}}$

5 $\mathbf{n + 4 = 10, \quad n = 6}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Simplify $6^3 \times 6^5$

SILVER Simplify $(a^2)^5 \div a^4$

GOLD Simplify $\dfrac{15x^6y^4}{3x^2y^3}$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $6^{3+5} = \mathbf{6^{8}}$

SILVER $a^{10} \div a^{4} = \mathbf{a^{6}}$

GOLD $5x^{6-2}y^{4-3} = \mathbf{5x^{4}y}$

Exam Practice — Multiplying and Dividing

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint $a^m \times a^n = a^{m+n}$ $\quad a^m \div a^n = a^{m-n}$

Question 1

(2 marks)

Simplify. Leave your answers in index form.

(a) (1)

$5^4 \times 5^3$

.....

(b) (1)

$7^9 \div 7^2$

.....

Question 2

(2 marks)

Write as a single power of 3

$$\frac{3^5 \times 3^2}{3^4}$$

.....

Exam Practice — Powers of Powers

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint $(a^m)^n = a^{mn}$ $a^0 = 1$

Question 3

(2 marks)

(a) Simplify (1)

$(x^3)^4$

.....

(b) Write down the value of (1)

9^0

.....

Question 4

(3 marks)

Find the value of n.

(a) (1)

$(2^3)^n = 2^{15}$

n = ...

(b) (2)

$5^n \div 5^2 = 5^6$

n = ...

Exam Practice — Simplifying Expressions

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint $a^m \times a^n = a^{m+n}$ $\quad a^m \div a^n = a^{m-n}$

Question 5

(2 marks)

Simplify fully

$$(3x^2y)^3$$

.....

Question 6

(2 marks)

Simplify fully

$$\frac{12a^5b^3}{4a^2b}$$

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 5^7 — B1

(b) 7^7 — B1

Tip: × add the powers, ÷ subtract the powers.

Question 2

$3^7 \div 3^4$ — M1

3^3 — A1

Tip: deal with the top first.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) x^{12} B1

(b) 1 B1

Tip: anything (except 0) to the power 0 is 1.

Question 4

(a) 3n = 15, n = 5 B1

(b) n − 2 = 6 M1

n = 8 A1

Tip: equal bases ⇒ equal powers.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$3^3 \times x^6 \times y^3$ — M1

$27x^6y^3$ — A1

Tip: cube the number AND every letter.

Question 6

$12 \div 4 = 3$ — M1

$3a^3b^2$ — A1

Tip: numbers divide, powers subtract.